



integrata
cegos

NoSQL

Databases for big and fast data

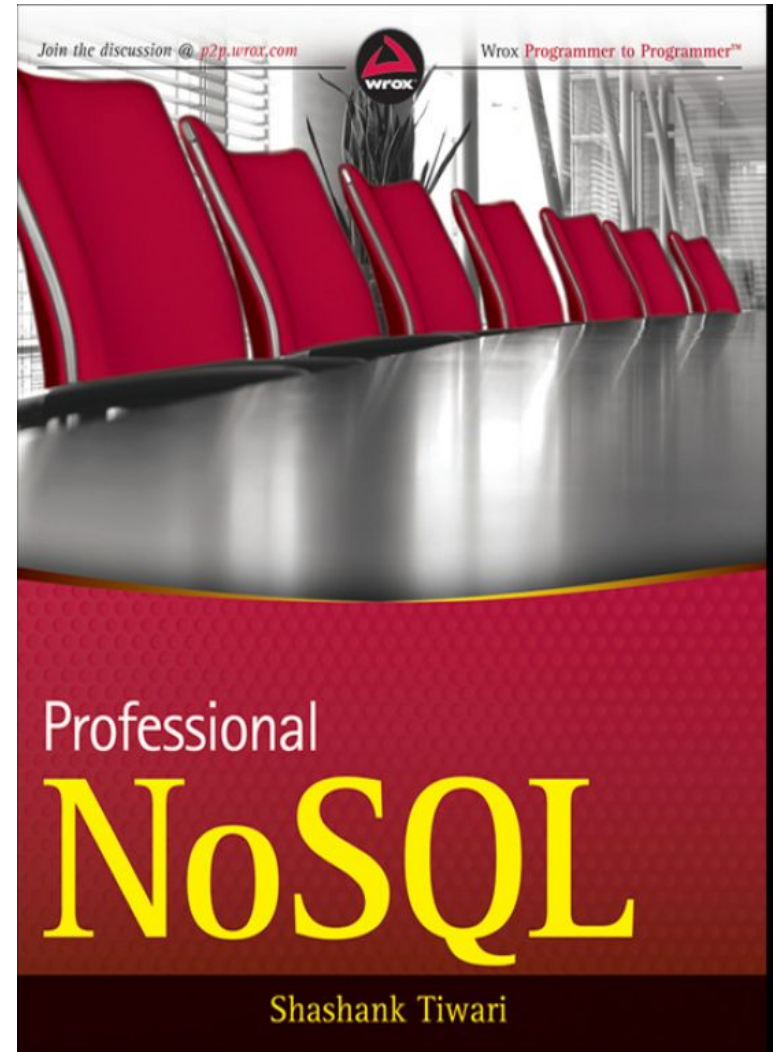
The
Pragmatic
Programmers

Seven Databases in Seven Weeks

A Guide to Modern Databases
and the NoSQL Movement

Eric Redmond
and Jim R. Wilson

Edited by Jacquelyn Carter



- The databases used are either open source or are used in the community edition
- Other tools used in this seminar are Open Source
 - LPGL license model
- A variety of products and technologies are used
 - Consoles
 - Query languages
 - Programming languages
- Programming
 - This allows the content to be deepened and internalized through exercises
 - A sample solution is provided in electronic form
 - However, this need not be the most elegant or best solution!

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THE PROBLEM

1.1

BIG & FAST DATA

What is "big"?

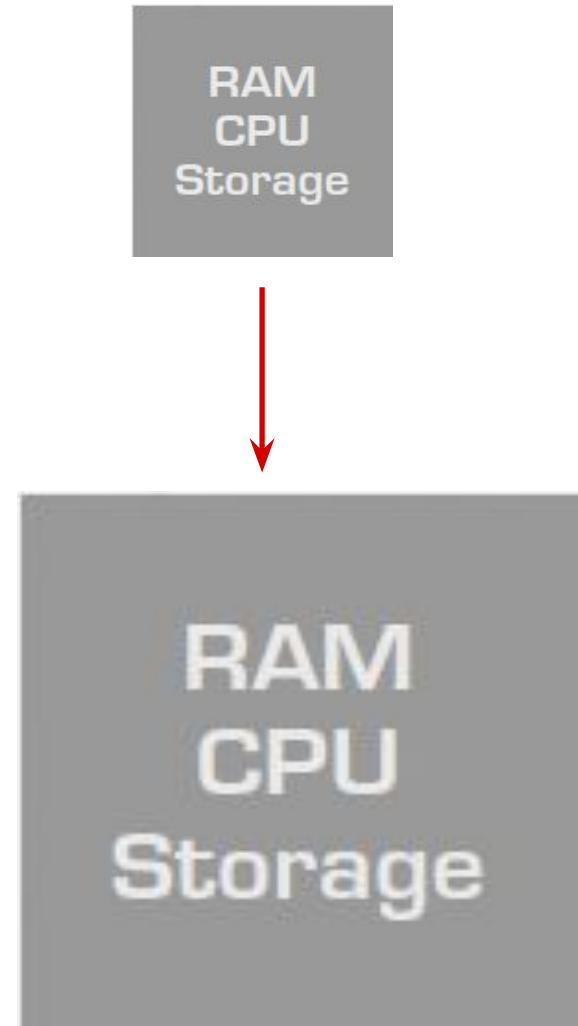
- Attempts at definition:
 - Physical
 - Terabyte
 - System
 - Exceeds the requirements of a normal storage medium
 - Technical
 - Data processing begins to cause problems due to size
- However:
 - No really clear definition recognizable
 - Boundaries can shift

Where do Big Datas occur?

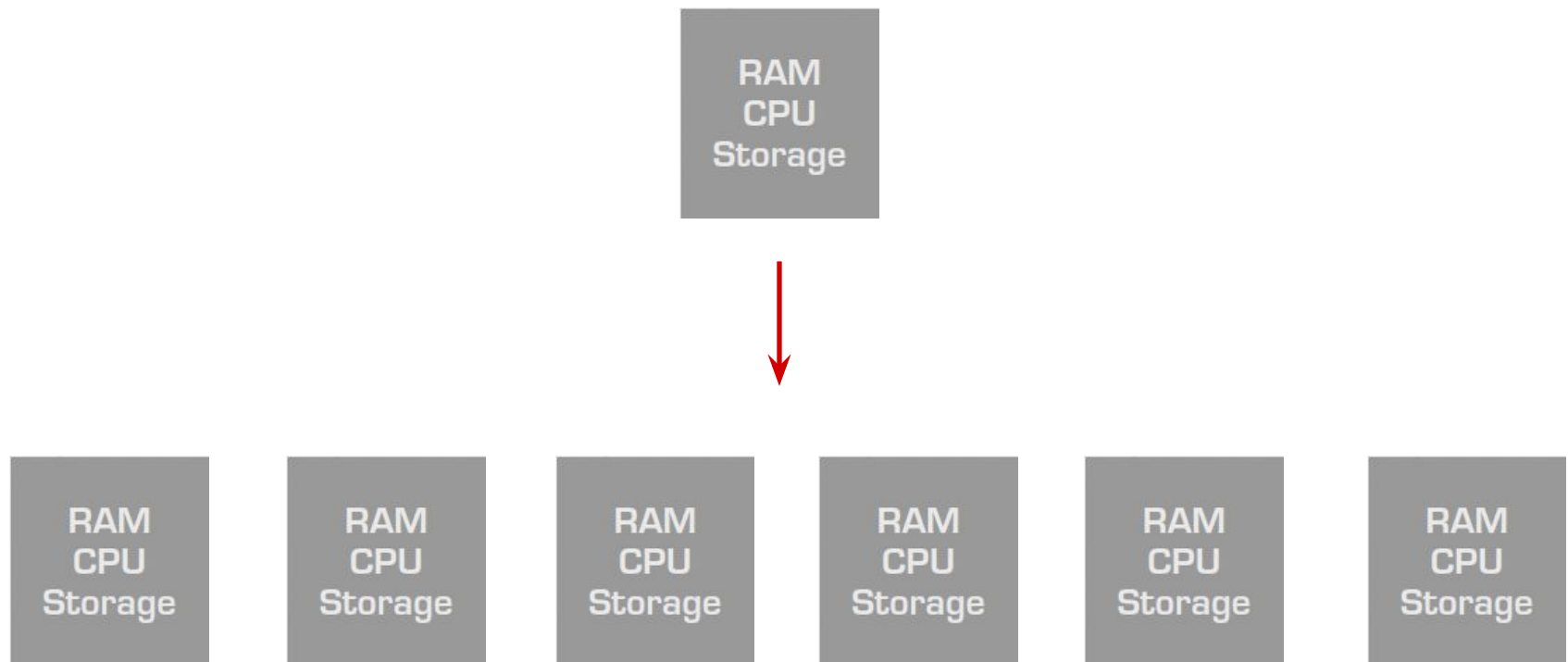
- Completely different areas:
 - DNS server for Internet
 - Indexing web pages for search engines
 - Google
 - Monitoring of page accesses, but also metrics information in large server systems
 - Content management and merchandise systems
 - Amazon
 - Social Media
 - Facebook

- Even powerful server machines are overwhelmed in the long run when it comes to storing and analyzing data
- Data storage is therefore distributed across several servers
- Load distribution is handled by a simple load balancer or a complex "master"
 - From the client's point of view, the cluster still appears as a single unit
- Optimally, the cluster is therefore a virtually infinite data sink with arbitrarily scalable computer capacity
 - A precursor to the "cloud"

- Boundaries are clearly defined
 - Costs
 - Current hardware limits
 - Fail-safe requirements



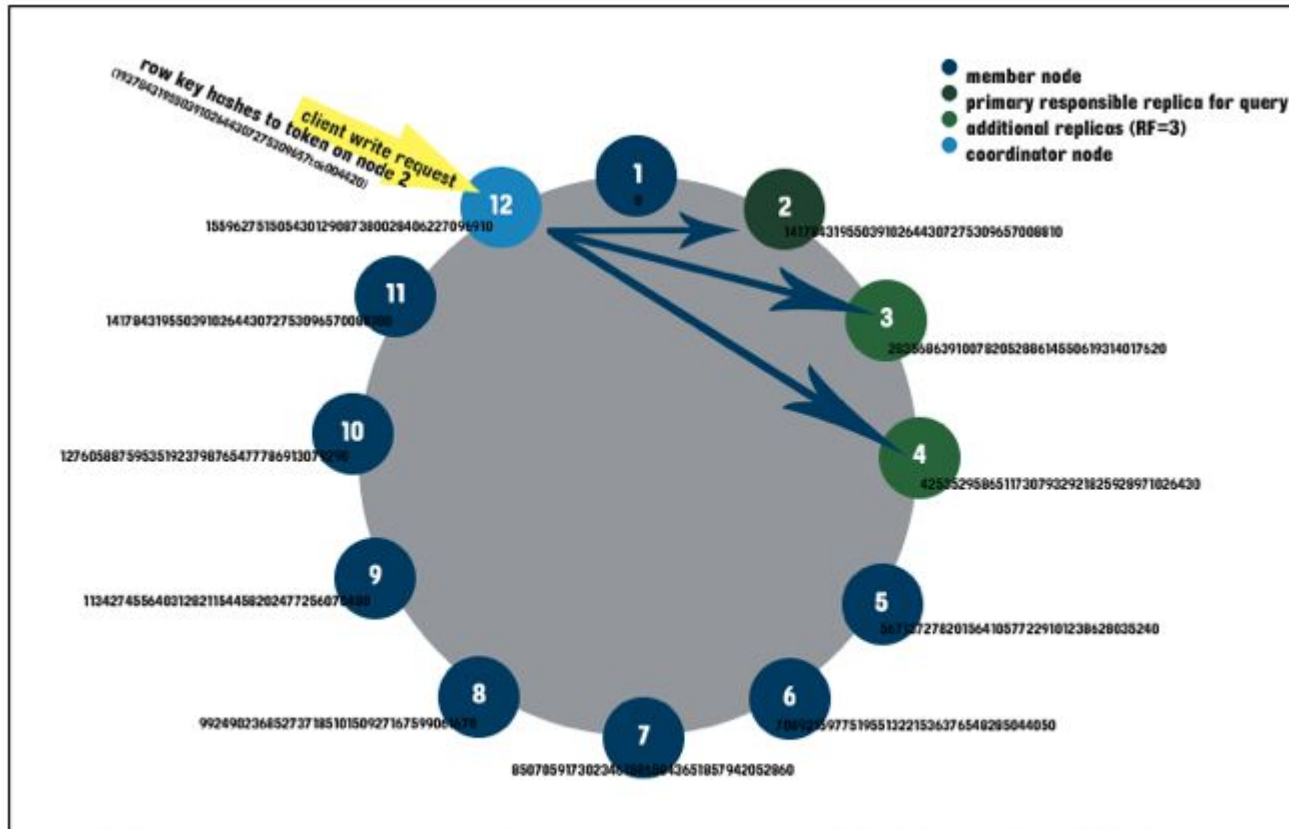
- Boundaries
 - Reliability requires replication via network
 - System administration and monitoring



The horizontally scaling ring cluster

- The data is distributed to a "ring" of individual servers based on unique hashes
- There is no master here
 - Each node can delegate to the server that actually holds the data

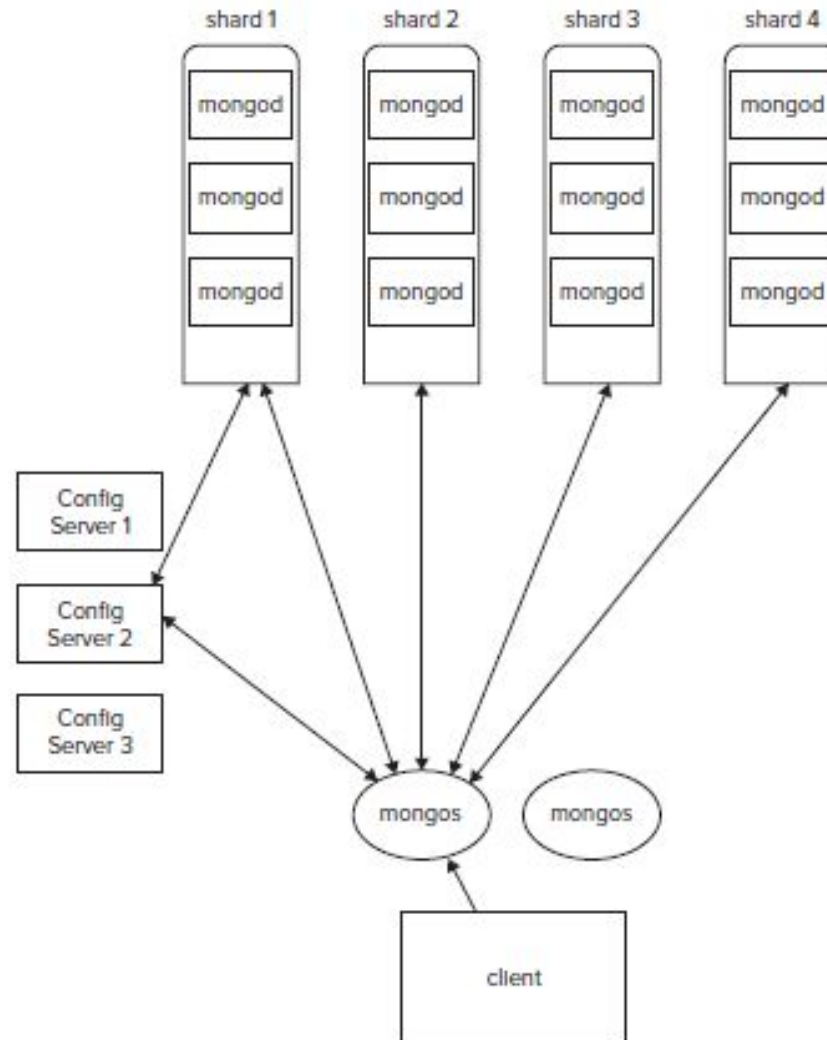
Example: Apache Cassandra Ring



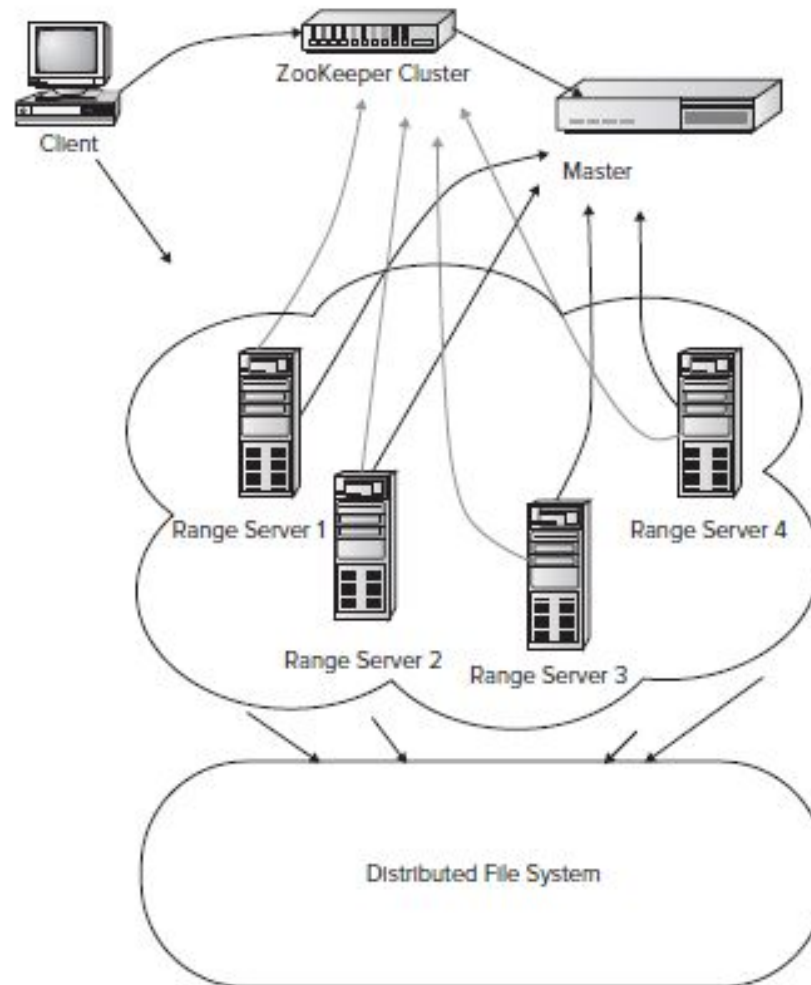
Quelle: <http://www.planetcassandra.org/blog/node-of-the-rings-fellowship-of-the-clusters-or-understanding-how-cassandra-stores-data/>

- Data is distributed to several computers in a sensible, well-defined manner
- A particular challenge arises in cluster operation
 - Adding a node can trigger "resharding"
 - This can lead to a considerable load on the server
 - Administrative process!

Example: Sharding and the Mongo DB



Example: Apache Hadoop

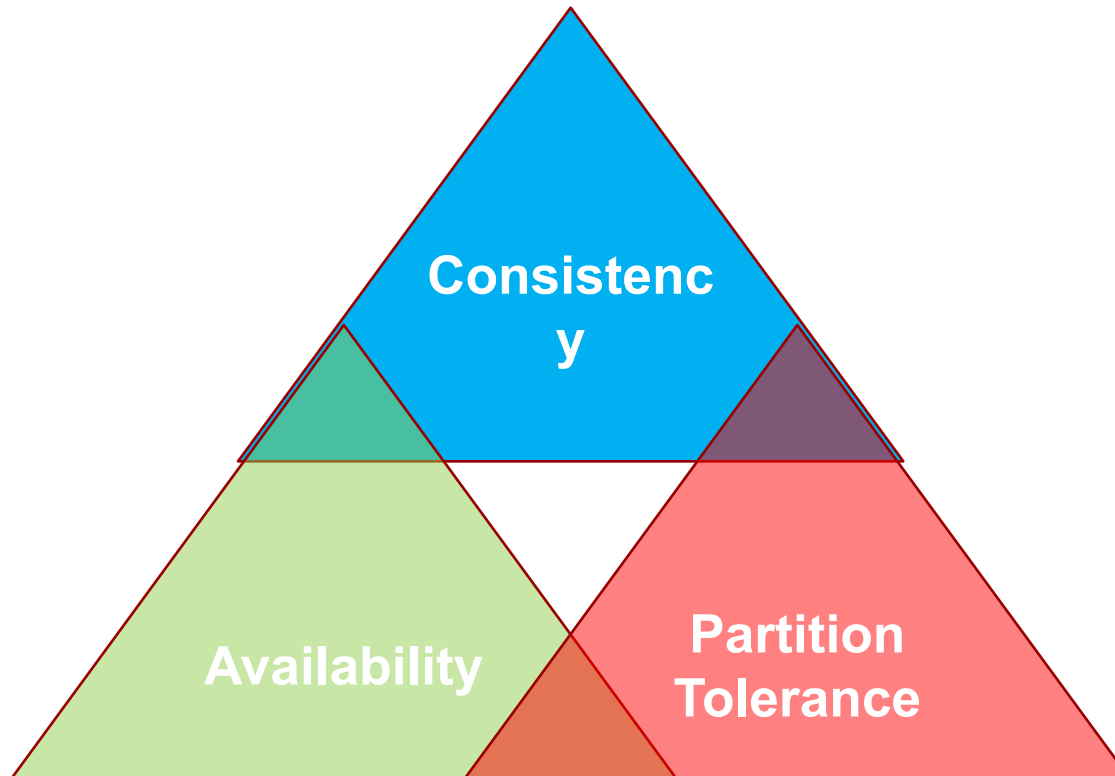


1.2

THE CAP THEOREM

- Consistency
 - All nodes have the same level of information at all times
- Availability
 - Every client is guaranteed a response
 - As long as at least one node is running in the cluster
- Partition Tolerance
 - The system tolerates
 - The failure of nodes
 - The failure of the network between nodes

No common intersection!



- Consistency
 - All nodes have the same level of information at **all times**
- Eventually Consistent
 - Changes to the database are propagated to the other nodes with a time delay
- BASE
 - Basically Available
 - Soft state
 - Eventual consistency

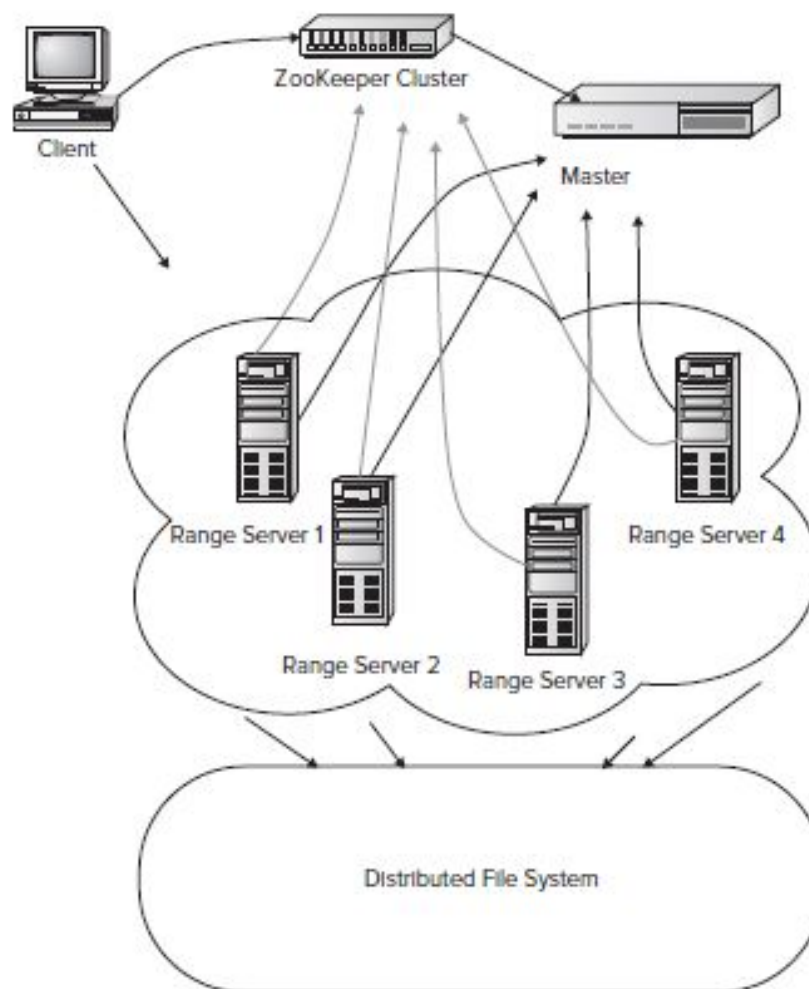
- ACID
 - Strong consistency of data
 - Transaction management
 - Transaction isolation
 - Two-phase commit if required
 - Relatively complex development
- BASE
 - Weak consistency
 - High availability
 - Speed
 - Fire-and-forget" if required
 - Easier development

1.3

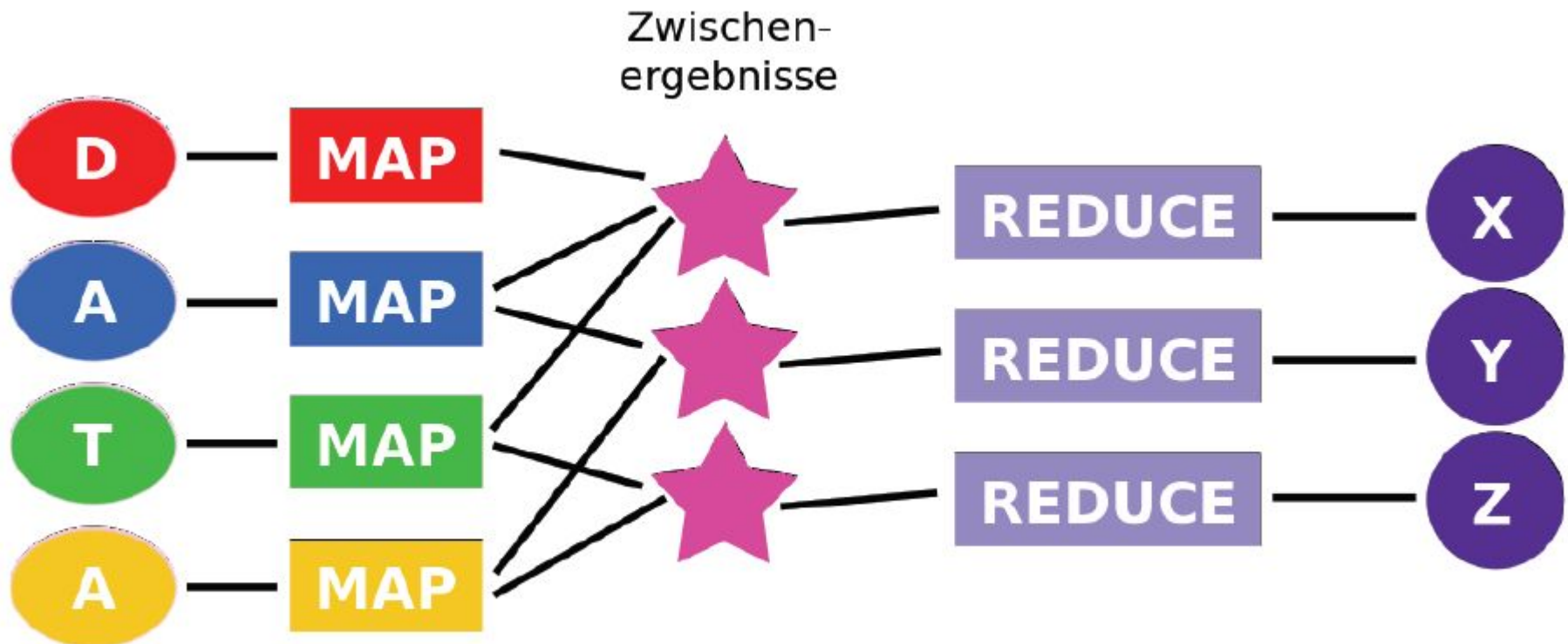
ANALYSIS OF BIG DATA

- Demands "new" algorithms
 - Breakdown into jobs
 - These are parallelized
 - Operating on a distributed database
- All of the companies mentioned have developed or used algorithms in the last 10 years
 - And "kindly" made available to the open source community
- Fast Data
 - Quick access to data selections
 - Fast implementation of new query requirements

Example: Apache Hadoop



- Groundbreaking development
 - Idea is attributed to Google
- Basic mode of operation
 - Operates on quantities (maps)
 - The first step is to prepare the map data in a new map
 - This is summarized in one or more Reduce steps to form the result structure



Quelle: <http://de.wikipedia.org>

1.4

SOME CLARIFICATIONS

NoSQL is not a complete package!

- NoSQL products require
 - Network infrastructure
 - Network
 - Router/Load Balancer
 - File system
 - Monitoring
 - Administration
 - System design

Hadoop is not NoSQL!

- Hadoop is used by some NoSQL products
 - Or can be used
- Is part of the infrastructure
 - Job Manager
 - Distributed File System

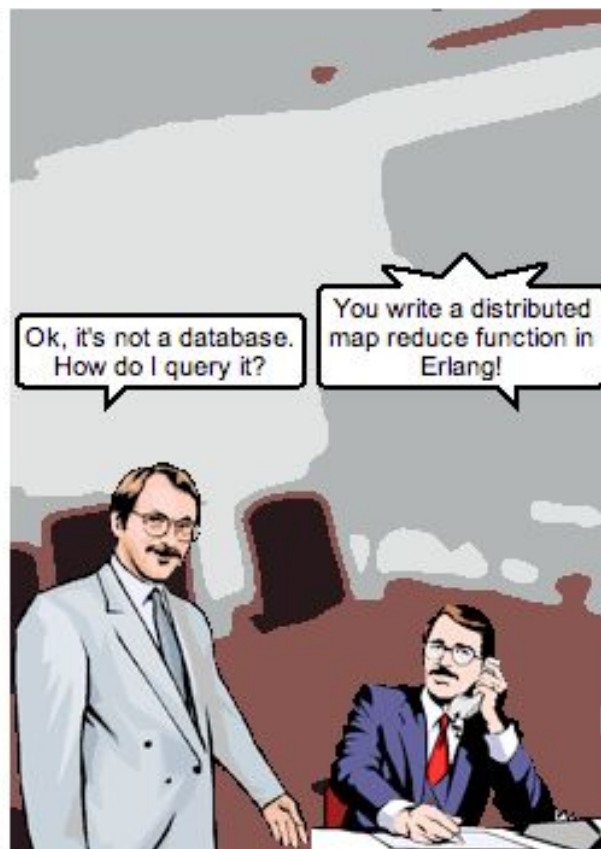
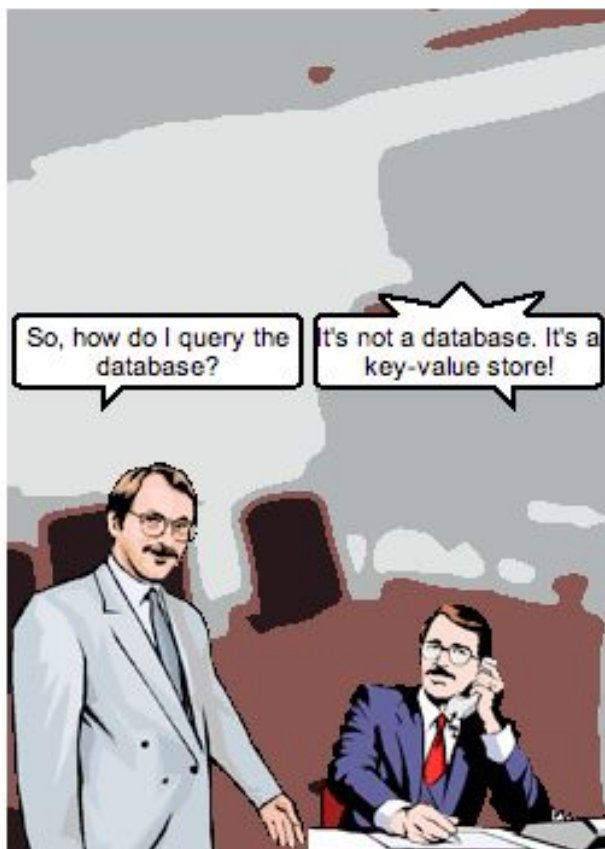
- Some parts of the history of NoSQL originate from problems with web applications
 - Session Management
 - DNS server
 - Content repositories
- However, many other areas of responsibility are unrelated to this
 - Content Management Systems
 - Caches
 - Tagging/indexing of big data
 - ...

NoSQL databases require the use of scripting languages

- Due to the lack of a static schema, mapping NoSQL structures into a statically compiled programming language is actually difficult
 - But not impossible
- Scripting languages such as Ruby, Python or JavaScript therefore usually generate much more compact programs
 - But you definitely don't have to use these languages
 - Mappings to Java, C# etc. can be used
- Many NoSQL databases make it possible to formulate MapReduce algorithms using JavaScript, for example
 - However, this is always optimized on the server side
 - There are also native alternatives
- The JSON format, which originates from JavaScript, has developed independently as an almost standard for data transfer

- Processing unstructured data in a NoSQL can be a great strength
 - "I don't need to know in detail how something is structured if I'm only interested in certain parts"
 - Exact format of a log message
 - Evaluation of server statistics for various products
 - Agile software development
- But NoSQL can also provide structure
 - Use of XML schema
 - Mandatory properties of a document

NoSQL is simple



- MapReduce functions
 - Are often defined as JavaScript
 - The program logic of a MapReduce function is very straightforward!
- Data format
 - JSON has established itself here
- Client access
 - RESTful Web Services

2

NOSQL

2.1

DEFINITION OF

- No SQL!
 - Original meaning
 - Somewhat dogmatic rejection of relational database principles
- Not only SQL
 - Already significantly weakened
 - Reduction to suitable problems
- No/ Not only relational
 - The actually correct term
 - Focus on data modeling, not on the query language

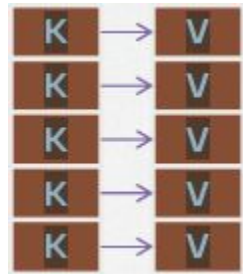
2.2

CLASSIFICATION

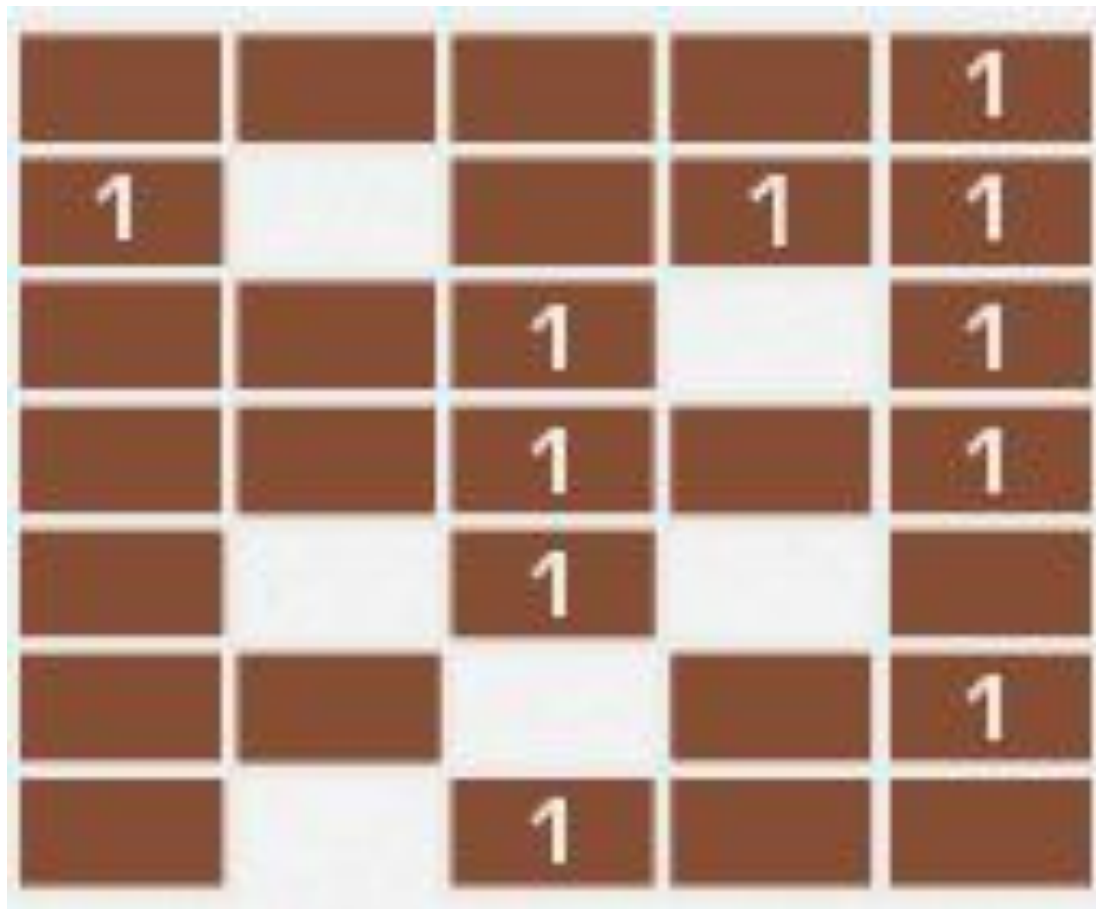
- Key/Value
- Column-oriented
- Document-oriented
- Graph-oriented

- Storage of content (=value) under a unique key
 - Flat structure
 - Values are viewed as a byte array
- Partial waiver of persistence of data
 - Cache systems
- Design for a large number of competing accesses

Key/Value: Mode of operation

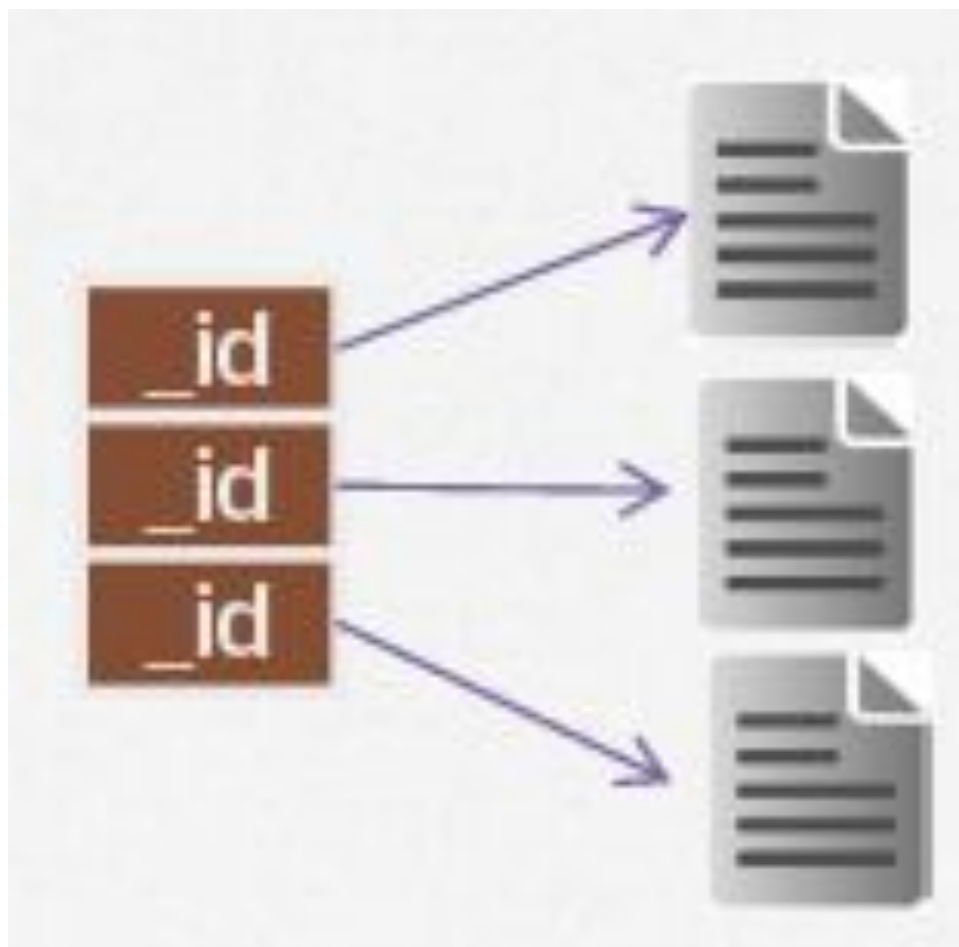


- Data is stored in columns
 - However, a row does not necessarily consist of all columns
- These columns can be further subdivided into groups
 - Advantageous for distributed data storage
- Focus on really large amounts of data



1				1
1			1	1
		1		1
		1		1
		1		
				1
		1		

- Storage of data in structured documents
 - Not necessarily XML!
 - JSON is much more widespread
 - Or the binary BSON



- Storage of data in
 - Node
 - Relationships
 - Attributes
- Nodes and relationships have attributes
- Nodes are linked to relationships and vice versa
- "Whiteboard friendly"
 - "What you can draw on a whiteboard you can store in a Graph DB."



2.3

DATA MODELING

- Data is modeled in an abstract way
 - Class-oriented programming
 - Java, C++/C#, ...
 - Table schemas
 - XML schema
- A conversion of data types requires recopying
 - "Migration"
- This means that development cycles and times for the introduction of changed features are relatively long

- "Duck typing": Only existing properties and implemented behavior count
 - "What is yellow, quacks and waddles is a duck"
- This is primarily the domain of dynamic scripting languages
 - Ruby, Python, JavaScript
- Data only needs to be copied in exceptional cases
- This supports agile software development

- What is...
 - Better?
 - Right?
 - Comfortable?
- Unfortunately no general answer possible
 - "Whoever leaves the room alive has won"
- But:
 - The trend in recent years has been increasingly towards Dynamil

3

PRODUCTS

3.1

OVERVIEW



- **Mongo DB**
 - A document-oriented database
- **Riak**
 - OneKey/Value-Store
- **Neo4J**
 - A graph-oriented database
- **HBase**
 - Column-oriented
- The respective documentation is available from the manufacturers on the web
 - Technical documentation
 - Tutorials and examples